

Supplementary Information

DYNAMIC MODELING FOR MULTICRITERIA DECISION SUPPORT IN THE END-OF-LIFE MANAGEMENT OF PHOTOVOLTAIC MODULES

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S1. SCOPE OF THE AHP APPLICATION

This material presents the calculation record used in the conceptual application of the Analytic Hierarchy Process (AHP). The objective is to make transparent how the weights reported in the manuscript were obtained, allowing the procedure to be reproduced, reviewed, or replaced by expert judgments in future applications. Scores were assigned based on the literature discussed in the manuscript, on the technical arguments associated with end-of-life routes, and on Saaty's fundamental scale. For order-3 matrices, $RI = 0.58$ was adopted in the consistency ratio calculation.

To guide the quantification of comparisons among criteria and alternatives, the scale shown in Table 1 was used. This scale translates qualitative arguments into numerical values compatible with AHP. When preference occurs in the opposite direction, the reciprocal value of the assigned score is used.

Table 1. Suggested scale for quantifying comparisons among criteria.

Weight	Importance
1	Equal importance between variables
2	Intermediate value between 1 and 3
3	Small or moderate importance of one variable over another
4	Intermediate value between 3 and 5
5	Strong or essential importance of one variable over another
6	Intermediate value between 5 and 7
7	Very strong or demonstrated importance of one variable over another
8	Intermediate value between 7 and 9
9	Absolute importance of one variable over another

The matrices were built in four blocks. First, the three main dimensions were compared: economic, environmental-social, and technical-operational. Then, the alternatives reuse, recycling, and disposal were compared separately within each dimension. Finally, the local weights of the alternatives were multiplied by the dimension weights, generating the final contribution of each route to the global ranking.

S2. COMPARISON AMONG DIMENSIONS

To define the values of level 2 of the hierarchy, the economic, environmental-social, and technical-operational dimensions were compared. As the manuscript emphasizes the construction of the economic indicator, this dimension received relative priority in the main matrix. This choice does not imply disregarding environmental impacts or operational feasibility; it represents the demonstrative scenario adopted to show how costs, revenues, avoided benefits, and externalities can guide decision support.

In the comparison between the economic dimension and the environmental-social dimension, a score of 3 was assigned in favor of the economic dimension. This judgment reflects that transportation, processing, and residue disposal costs can dominate the feasibility of EoL strategies, although the environmental motivation remains central to avoiding photovoltaic waste accumulation and material losses (Farrell et al. 2020; Nain e Kumar 2022; Perkins et al. 2014; Santos e Ogunseitan 2022). Thus, the score 3 represents a moderate preference for the economic dimension in the analyzed scenario.

In the comparison between the economic dimension and the technical-operational dimension, a score of 5 was assigned in favor of the economic dimension. The rationale is that the study seeks to demonstrate the formulation of the economic indicator and its effects on ranking; in addition, infrastructure, inspection, processing, and disposal are translated into costs that directly affect the decision (Brito et al. 2021; Ho et al. 2025; Yu et al. 2022). The technical-operational dimension remains relevant, but appears as a condition for the economic feasibility of the routes.

Finally, the environmental-social and technical-operational dimensions were considered equally important, resulting in a score of 1. This equivalence indicates that the expected environmental benefit depends on technical execution capacity: reuse without testing and traceability may generate risks, whereas recycling without adequate technology may produce secondary waste and reduce effective circularity (Farrell et al. 2020; Ho et al. 2025; Latunussa et al. 2016; Marinna et al. 2025; Markert et al. 2020).

Table 2 presents the pairwise comparison matrix of the criteria, and Table 3 presents the normalized matrix, obtained weights, and consistency indicators.

Table 2. Pairwise comparison matrix of the criteria.

	Economic	Environmental-social	Technical-operational
Economic	1	3	5
Environmental-social	0.33333	1	1
Technical-operational	0.2	1	1

Table 3. Normalized matrix of the criteria and weights.

	Economic	Environmental-social	Technical-operational	Weight
Economic	0.6522	0.6000	0.7143	0.6555
Environmental-social	0.2174	0.2000	0.1429	0.1867

Technical-operational	0.1304	0.2000	0.1429	0.1578
	$\lambda_{Max} = 3,0432$	CI = 0.0216	CR = 3,72%	

S3. ECONOMIC DIMENSION

The economic dimension considers the capacity of each route to recover value or reduce net costs. In this stage, the alternatives were compared based on potential revenues, decommissioning costs, transportation, inspection, reconditioning, processing, residue disposal, and monetizable externalities. Reuse received the highest preference because it can generate resale revenue and avoid processing costs. Recycling was considered preferable to disposal because it recovers materials, although it involves transportation, processing, and residue costs. Disposal was treated as a net cost, with no value recovery.

Reuse was considered economically favorable when a secondary market exists and modules require little repair. Under this condition, the financial benefit is obtained from resale revenue, discounted by logistics, inspection, testing, and reconditioning costs (Marinna et al. 2025). This route also avoids, at least temporarily, costs associated with full module processing.

Recycling presents a decision paradox: it recovers materials with economic value, such as aluminum, glass, silicon, and metals, but has relevant transportation, processing, and residue disposal costs. Its feasibility depends on operational scale, material separation efficiency, and the existence of markets for reinserting recovered fractions (Brito et al. 2021; Markert et al. 2020).

Disposal corresponds to the route of lowest immediate operational complexity and may present lower direct short-term cost in some scenarios. However, it should be interpreted as a total net cost because it generates no revenue, recovers no material value, and may involve landfill fees, pretreatments, and transfer of environmental costs to the future (Markert et al. 2020).

In the pairwise comparisons, a score of 5 was assigned to reuse over recycling, indicating strong importance of reuse in the economic dimension because this route can generate direct revenue with less need for processing. The comparison between reuse and disposal received a score of 7 because reuse transforms the module into a marketable asset, whereas disposal represents a net cost. Recycling received a score of 3 over disposal because, although it may be operationally more expensive, it recovers material value and avoids total resource loss.

Table 4 presents the pairwise comparison matrix of the subcriteria of Economic Dimension, and Table 5 presents the normalized matrix, obtained weights, and consistency indicators.

Table 4. Pairwise comparison matrix of the economic dimension.

	Reuse	Recycling	Disposal
Reuse	1	5	7

Recycling	0.2	1	3
Disposal	0.142	0.3333	1

Table 5. Normalized matrix of the economic dimension and weights.

	Reuse	Recycling	Disposal	Weight
Reuse	0.7447	0.7895	0.6364	0.7235
Recycling	0.1489	0.1579	0.2727	0.1932
Disposal	0.1064	0.0526	0.0909	0.0833
	$\lambda_{Max} = 3.1115$	CI = 0.0557	CR = 9.61%	

S4. ENVIRONMENTAL-SOCIAL DIMENSION

The environmental-social dimension considers avoided or generated waste, emissions, occupational risks, material loss, and social benefits. Reuse was prioritized because it postpones waste generation, reduces demand for new modules, and increases the use of second-life equipment. Recycling was considered superior to disposal because it recovers materials and reduces direct disposal, although it may generate secondary waste, emissions, and occupational risks. Disposal received the lowest priority because it represents material loss and transfer of environmental liabilities.

In this dimension, the analysis focuses on the balance between avoided impacts and generated impacts (Brito et al. 2021; Ho et al. 2025; Markert et al. 2020; Yu et al. 2022). Reuse is the route most aligned with prevention because it extends module service life and reduces the need to manufacture new equipment (Farrell et al. 2020; Nain e Kumar 2022). In addition, it may generate social benefit by expanding access to lower-cost modules in compatible second-life applications (Marinna et al. 2025).

Recycling contributes to recovering materials and reducing demand for primary resources. However, it does not eliminate all end-of-life impacts, as it may generate secondary waste, transport-related emissions, and occupational risks during processing (Markert et al. 2020). Therefore, although environmentally preferable to disposal, it was not considered superior to reuse in this dimension.

Disposal showed the worst environmental-social performance because it implies definitive material loss, lack of circularity, and possible transfer of environmental liabilities (Brito et al. 2021; Ho et al. 2025; Markert et al. 2020; Yu et al. 2022). This route also does not generate direct social benefits associated with life extension or material recovery.

In the pairwise comparisons, a score of 3 was assigned to reuse over recycling because reuse avoids or postpones the generation of secondary residues associated with processing. The comparison between reuse and disposal received a score of 9, indicating absolute importance of reuse, since this route extends product use and avoids immediate final disposal. Recycling received a score of 5 over disposal because it

recovers materials and reduces total resource loss, despite generating processing impacts.

Table 6 presents the pairwise comparison matrix of the subcriteria of environmental-social dimension, and Table 7 presents the normalized matrix, obtained weights, and consistency indicators.

Table 6. Pairwise comparison matrix of the environmental-social dimension.

	Reuse	Recycling	Disposal
Reuse	1	3	9
Recycling	0.33333	1	5
Disposal	0.1111	0.2	1

Table 7. Normalized matrix of the environmental-social dimension and weights.

	Reuse	Recycling	Disposal	Weight
Reuse	0.6923	0.7143	0.6000	0.6689
Recycling	0.2308	0.2381	0.3333	0.2674
Disposal	0.0769	0.0476	0.0667	0.0637
	$\lambda_{Max} = 3.0453$	CI = 0.0226	CR = 3.90%	

S5. TECHNICAL-OPERATIONAL DIMENSION

The technical-operational dimension considers route maturity, infrastructure availability, safety, traceability, and process quality (Brito et al. 2021; Ho et al. 2025; Markert et al. 2020; Yu et al. 2022). Recycling received the highest weight because it has industrial routes applicable to large volumes and depends on specialized plants already structured in some contexts. Reuse remained close to recycling, but strongly depends on visual inspection, electrical testing, traceability, certification, and safety assurance for a second life. Disposal was considered operationally simple, but hierarchically inferior because it does not promote circularity.

Reuse depends on verifying residual performance and physical integrity of the modules. To be technically safe, it requires visual inspection, electrical tests, insulation verification, I-V curve analysis, electroluminescence, traceability, and minimum certification or warranty criteria (Marinna et al. 2025). When these requirements are met, the route preserves the original function of the product and reduces pressure on processing and disposal.

Recycling corresponds to a technically more complex route, but one that is more suitable for large volumes of modules without use conditions. Its performance depends on specialized infrastructure, technological maturity, and quality of separation processes (Ho et al. 2025; Yu et al. 2022). The main technical challenge lies in delamination and efficient separation of glass, cells, metals, and polymers, so that recovered materials can be reinserted into the production chain (Brito et al. 2021; Ho et al. 2025; Yu et al. 2022).

Disposal is operationally simple in the short term, as it depends mainly on logistics and availability of licensed disposal systems (Markert et al. 2020). However, its technical-operational performance is lower because it does not use the functional or material potential of modules and does not contribute to circularity strategies.

In the pairwise comparisons, reuse and recycling were considered equivalent, receiving a score of 1. Reuse may be technically simpler when the module is intact, whereas recycling has industrial processes better suited to large volumes, although it faces separation challenges. The comparison between reuse and disposal received a score of 5 because reuse requires technical protocols and preserves the product function. Recycling received a score of 7 over disposal because it involves technological processes for material recovery and presents superior technical performance compared with mere landfilling.

Table 8 presents the pairwise comparison matrix of the subcriteria of technical-operational dimension, and Table 9 presents the normalized matrix, obtained weights, and consistency indicators.

Table 8. Pairwise comparison matrix of the technical-operational dimension.

	Reuse	Recycling	Disposal
Reuse	1	1	5
Recycling	1	1	7
Disposal	0.2	0.1429	1

Table 9. Normalized matrix of the technical-operational dimension and weights.

	Reuse	Recycling	Disposal	Weight
Reuse	0.4545	0.4667	0.3846	0.4353
Recycling	0.4545	0.4667	0.5385	0.4866
Disposal	0.0909	0.0667	0.0769	0.0782
	$\lambda_{Max} = 3.0164$	CI = 0.0082	CR = 1.41%	

S6. PAIRWISE COMPARISON MATRIX OF THE CRITERIA

The final aggregation combines the dimension weights with the local weights of the alternatives. For each route, the partial contribution is obtained by multiplying the dimension weight (α_d) by the local weight of the route in that dimension ($w_i^{(d)}$). The final score was obtained by $V_i = \sum_d \alpha_d w_i^{(d)}$, as shown in Table 10.

Table 10. Pairwise Comparison Matrix of the Criteria

Criterion (Weight)	Sub criterion (Weight)	Final Weight
Economic (0.6555)	Reuse (0.7235)	0.4742
	Recycling (0.1932)	0.1266
	Disposal (0.0833)	0.0546
Environmental-social (0.1867)	Reuse (0.6689)	0.1249
	Recycling (0.2674)	0.0499
	Disposal (0.0637)	0.0119
Technical-operational (0.1578)	Reuse (0.4353)	0.0687
	Recycling (0.4866)	0.0768
	Disposal (0.0782)	0.0123

Following this logic, reuse reached a final score of 0.6678, recycling obtained 0.2533, and disposal obtained 0.0788. Thus, the final ranking of the conceptual application is: reuse, recycling, and controlled disposal. This result should be interpreted as a methodological demonstration and may be changed in future applications by updating the weights, including experts, or replacing the scores with regional data.